

INFORMATION TECHNOLOGY INSTITUTE

LAB 8

Negative Feedback





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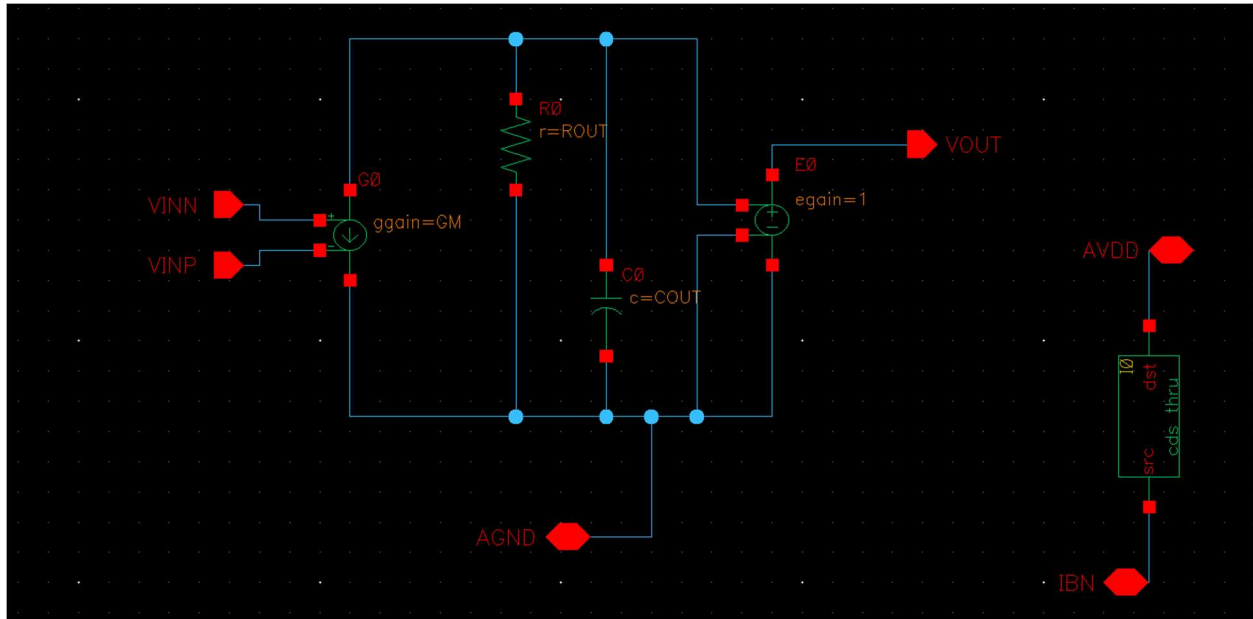
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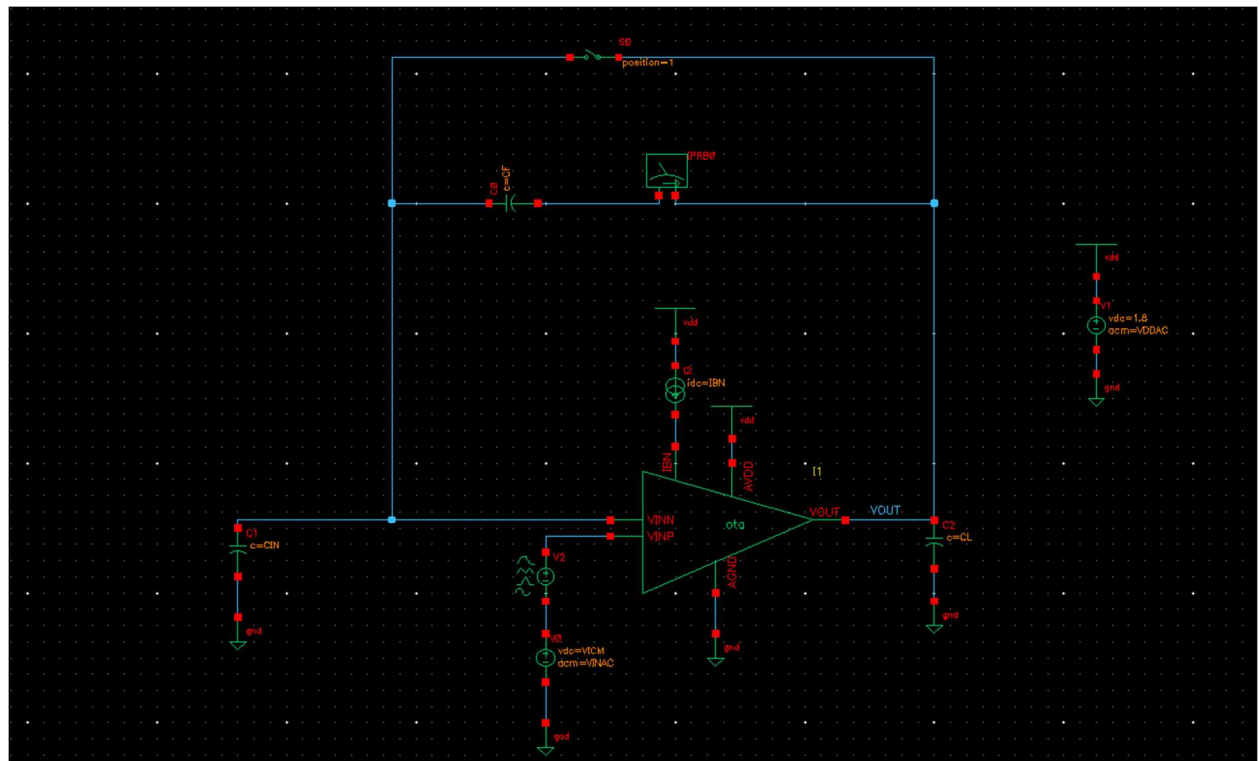


PART 1: Feedback with Behavioral OTA

Schematic



TB Schematic





Variables

Description	Variable	Value
Feedback capacitance	CF	4p
Input capacitance	CIN	4p, 12p
Load capacitance	CL	as in Lab 07
Transconductance of behavioral model	GM	as in Lab 07
DC gain of behavioral model	Av	as in Lab 07
UGF of behavioral model	wu ¹	as in Lab 07
Output cap of behavioral model	COUT	GM/wu
Output res of behavioral model	ROUT	Av/GM
AC stimulus magnitude	VINAC	1
Transient stimulus peak	VP	50m
Transient stimulus frequency	FIN	1k
Bias current	IB	as in Lab 07
CM input voltage	VICM	At the middle of the CMIR

Design Variables

GM

CF

CIN

CL

IBN

VDDAC

VICM

VINAC

WU

AV

ROUT

COUT

163.2u

4p

4p

5p

20u

0

1.35

1

31.417M

51

AV/GM

GM/WU

Click to add variable

Click to add test

Global Variables

CF

CIN

CL

IBN

VDDAC

VICM

VINAC

COUT

GM

WU

AV

ROUT

4p

4p

5p

20u

0

1.35

1

GM/WU

163.2u

31.417M

0

AV/GM

Click to add variable

1. Closed loop gain vs frequency

VOU

Sat Aug 15 14:03:36 2026 1

Name Vis V1 CIN

VOUT 5.68809dB 4e-12

VOUT 11.3884dB 1.2e-11

V (dB)

freq (Hz)

40.6342Hz

11.3884dB

5.68809dB

2.59064MHz 2.66523dB

4.28598MHz

4.83157MHz

1.33806MHz 8.39762dB

dx:1.25258MHz dy:5.73239dB s:4.57645u



Hand analysis

$$\beta = \frac{C_F}{C_F + C_{IN}}, A_{CL}(0) = \frac{A_v}{1 + \beta A_v}$$

$$BW_{CL} = \frac{\omega_u}{2\pi A_v} (1 + \beta A_v)$$

For $C_{in} = 4\text{p}$

$$A_{CL}(0) = \frac{51}{1 + (0.5)(51)} = 1.9245$$

$$BW_{CL} = \frac{31.417 \times 10^6}{2\pi(51)} [1 + (0.5)(51)] = 2.598 \text{ MHz}$$

For $C_{in} = 12\text{p}$

$$A_{CL}(0) = \frac{51}{1 + (0.25)(51)} = 3.7091 \text{ V/V}$$

$$BW_{CL} = \frac{31.417 \times 10^6}{2\pi(51)} [1 + (0.25)(51)] = 1.348 \text{ MHz}$$

$$GBW = A_{CL}(0) \times BW_{CL} = \frac{\omega_u}{2\pi}$$

$$GBW = \frac{31.417 \times 10^6}{2\pi} = 5.000 \text{ MHz}$$

Comparison

CIN	Parameter	Hand analysis	Simulation
4 pF	DC gain	1.9245 V/V	1.925 V/V
4 pF	DC gain (dB)	5.686 dB	5.688 dB
4 pF	Bandwidth	2.598 MHz	2.597 MHz
4 pF	GBW	5.000 MHz	5.012 MHz

Point	Test	Output	Nominal
Parameters: CIN=4p			
1	lab8:TB:1	AVdb	5.688
1	lab8:TB:1	AV	1.925
1	lab8:TB:1	BW	2.597M
1	lab8:TB:1	GBW	5.012M
Parameters: CIN=12p			
2	lab8:TB:1	AVdb	11.39
2	lab8:TB:1	AV	3.71
2	lab8:TB:1	BW	1.349M
2	lab8:TB:1	GBW	5.018M

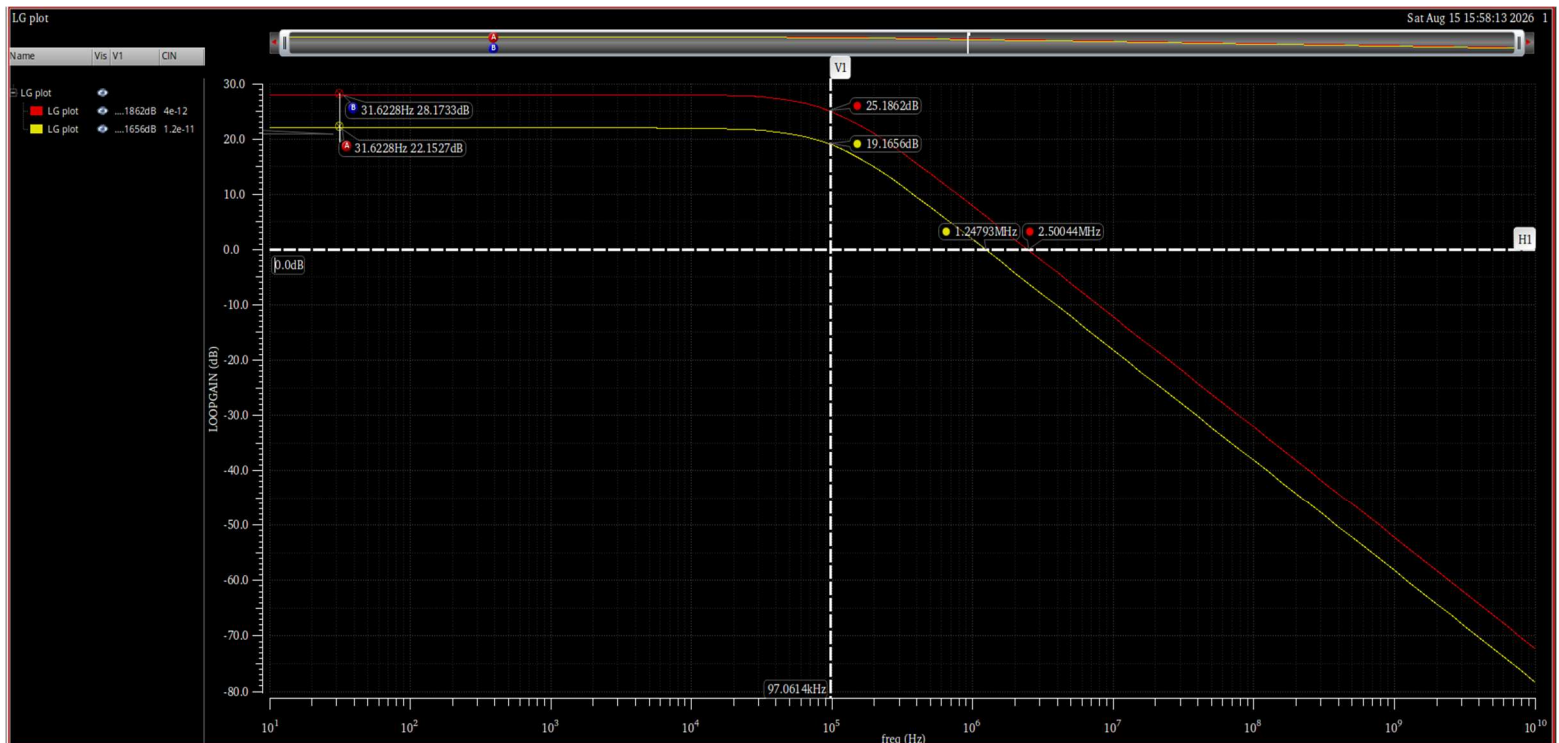


CIN	Parameter	Hand analysis	Simulation
12 pF	DC gain	3.7091 V/V	3.710 V/V
12 pF	DC gain (dB)	11.385 dB	11.390 dB
12 pF	Bandwidth	1.348 MHz	1.349 MHz
12 pF	GBW	5.000 MHz	5.018 MHz

Comment on the difference between the results for the two values of CIN

As The cin increase from 4p to 12p the Acl increase form 2 to 4 ideally and the BW decreases to half ,however the GBW stayed constant and this is because the $BW_{cl} = \beta \cdot W_u$ and W_u is constant as it doesn't depend on the R_{out} it is g_m / C_{out} .

Loop gain vs frequency





Hand analysis

$$\beta = \frac{C_F}{C_F + C_{IN}}, \quad LG(0) = \beta A_v$$

$$BW_{LG} = \frac{\omega_u}{2\pi A_v}, \quad GBW_{LG} = LG(0) \times BW_{LG}$$

For $C_{in} = 4p$

$$\beta = \frac{4}{4+4} = 0.5, \quad LG(0) = (0.5)(51) = 25.5 \text{ V/V}$$

$$LG_{dB}(0) = 20 \log_{10}(25.5) = 28.13 \text{ dB}$$

$$BW_{LG} = \frac{31.417 \times 10^6}{2\pi(51)} = 98.04 \text{ kHz}$$

$$GBW_{LG} = (25.5)(98.04 \text{ kHz}) = 2.500 \text{ MHz}$$

For $C_{in} = 12p$

$$\beta = \frac{4}{4+12} = 0.25, \quad LG(0) = (0.25)(51) = 12.75 \text{ V/V},$$

$$LG_{dB}(0) = 20 \log_{10}(12.75) = 22.11 \text{ dB}$$

$$BW_{LG} = \frac{31.417 \times 10^6}{2\pi(51)} = 98.04 \text{ kHz}$$

$$GBW_{LG} = (12.75)(98.04 \text{ kHz}) = 1.250 \text{ MHz}$$

Point	Test	Output	Nominal
Parameters: CIN=4p			
1	lab8:TB:1	LG UGF	2.502M
1	lab8:TB:1	LG BW	97.47k
1	lab8:TB:1	LG	25.62
1	lab8:TB:1	LGdb	28.17
Parameters: CIN=12p			
2	lab8:TB:1	LG UGF	1.249M
2	lab8:TB:1	LG BW	97.47k
2	lab8:TB:1	LG	12.81
2	lab8:TB:1	LGdb	22.15

Comparison

CIN	Parameter	Hand analysis	Simulation
4 pF	DC loop gain	25.50 V/V	25.62 V/V
4 pF	DC loop gain (dB)	28.13 dB	28.17 dB
4 pF	GBW	2.500 MHz	2.502 MHz

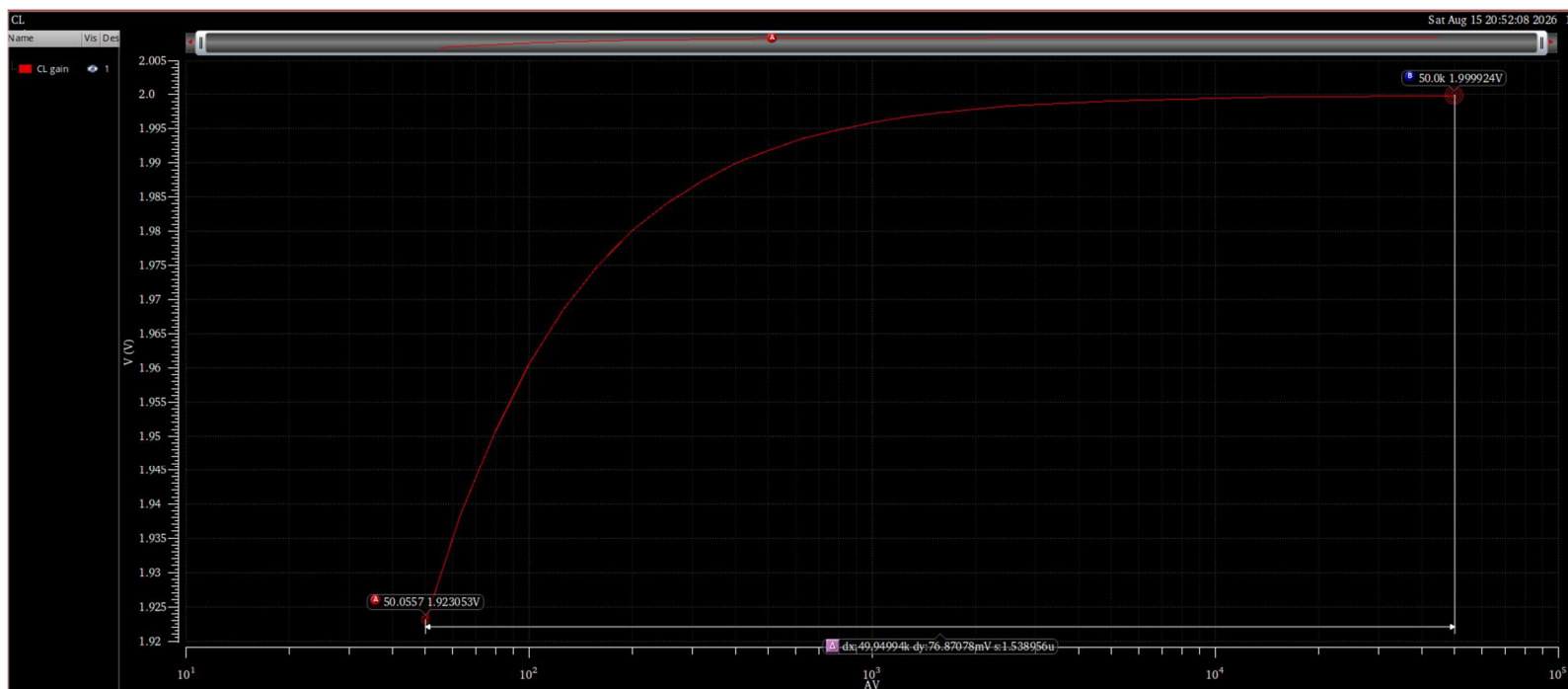


CIN	Parameter	Hand analysis	Simulation
12 pF	DC loop gain	12.75 V/V	12.81 V/V
12 pF	DC loop gain (dB)	22.11 dB	22.15 dB
12 pF	GBW	1.250 MHz	1.249 MHz

Comment

As beta decreases the Acl increase as Acl is approximately $1/\beta$ and the UGF increases as a result of shifting the curve as it is being multiplied by a smaller value but the LG poles stay at the same frequency as beta is a real number.

Gain Desensitization





Percentage of change

lab8:TB:1	max CL gain	2
lab8:TB:1	min CL gain	1.923
lab8:TB:1	percentage of change	3.996

Comment

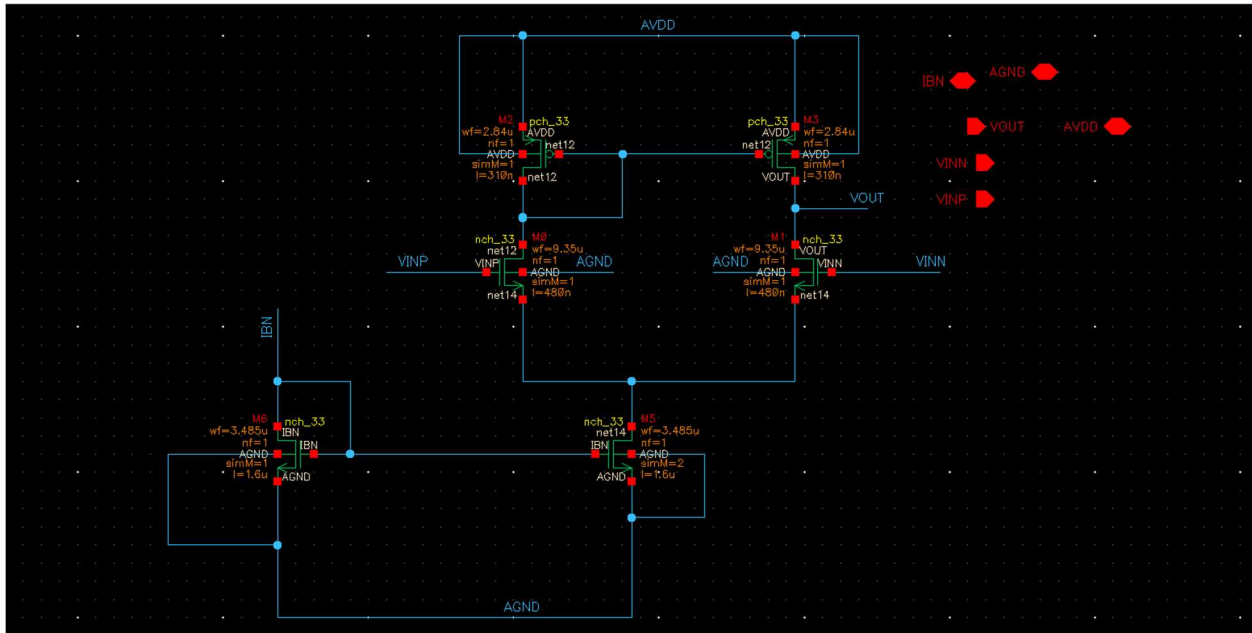
As the Aol increase the static error decrease as

$\varepsilon_s = \frac{A_{CL,ideal} - A_{CL}^{(0)}}{A_{CL,ideal}} \times 100\%$ since $A_{CL,ideal} = \frac{1}{\beta}$ $\varepsilon_s = \frac{1}{1 + \beta A_v} \times 100\%$ which is approximately $1/LG$ so as the Aol increase the LG increase thus decreasing the static error

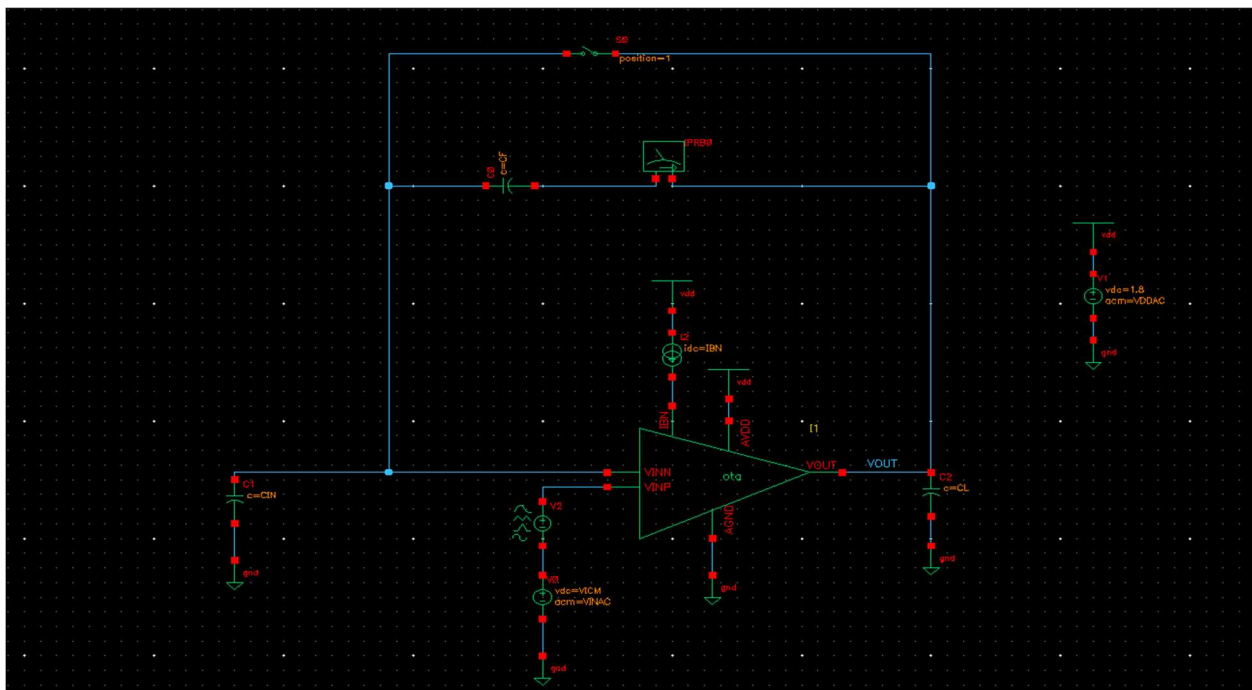


PART 2: Feedback with Real 5T OTA

Schematic

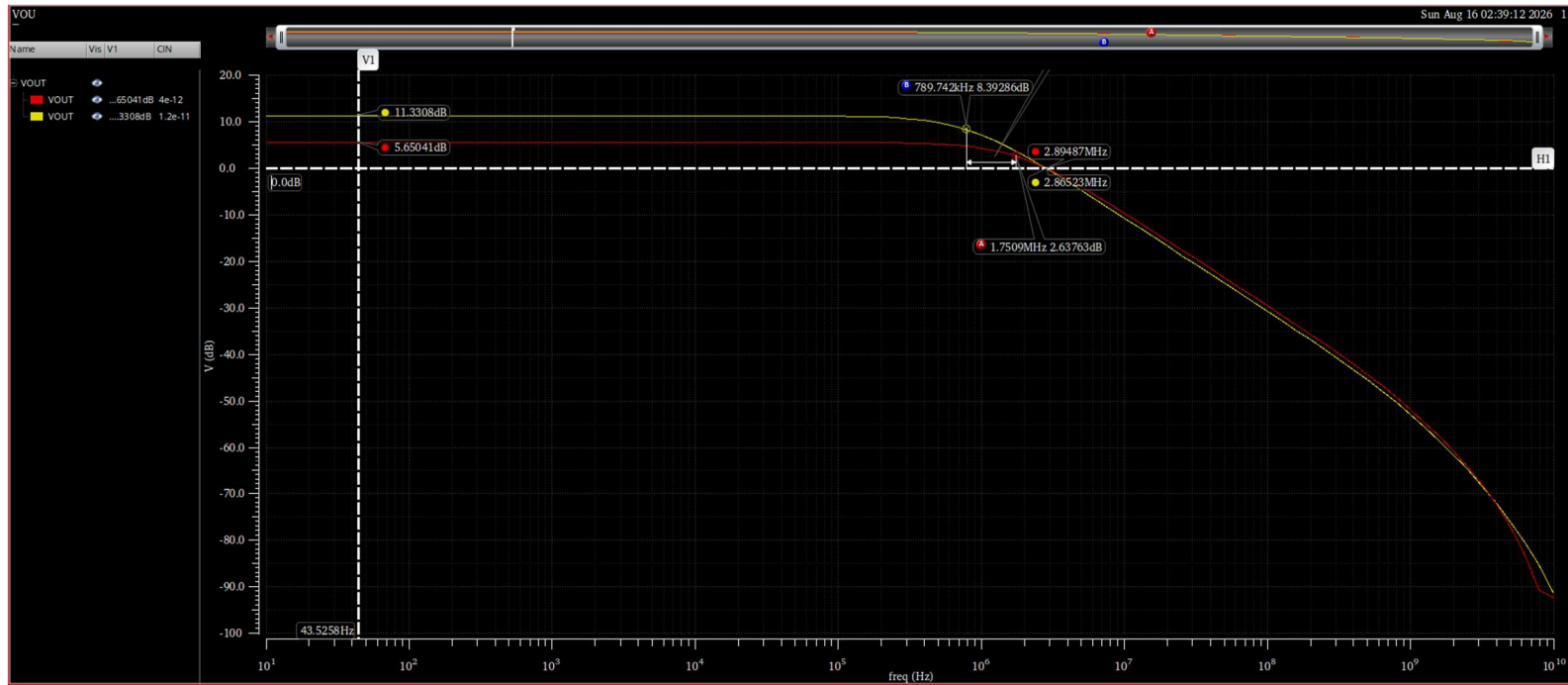


TB Schematic





Closed loop gain vs frequency



Comparison

CIN	Parameter	Part 1: Behavioral OTA	Part 2: Real 5T OTA
4 pF	DC gain	1.925 V/V	1.917 V/V
4 pF	DC gain (dB)	5.688 dB	5.650 dB
4 pF	Bandwidth	2.597 MHz	1.767 MHz
4 pF	GBW	5.012 MHz	3.395 MHz
12 pF	DC gain	3.710 V/V	3.686 V/V
12 pF	DC gain (dB)	11.390 dB	11.330 dB
12 pF	Bandwidth	1.349 MHz	801.8 kHz
12 pF	GBW	5.018 MHz	2.963 MHz

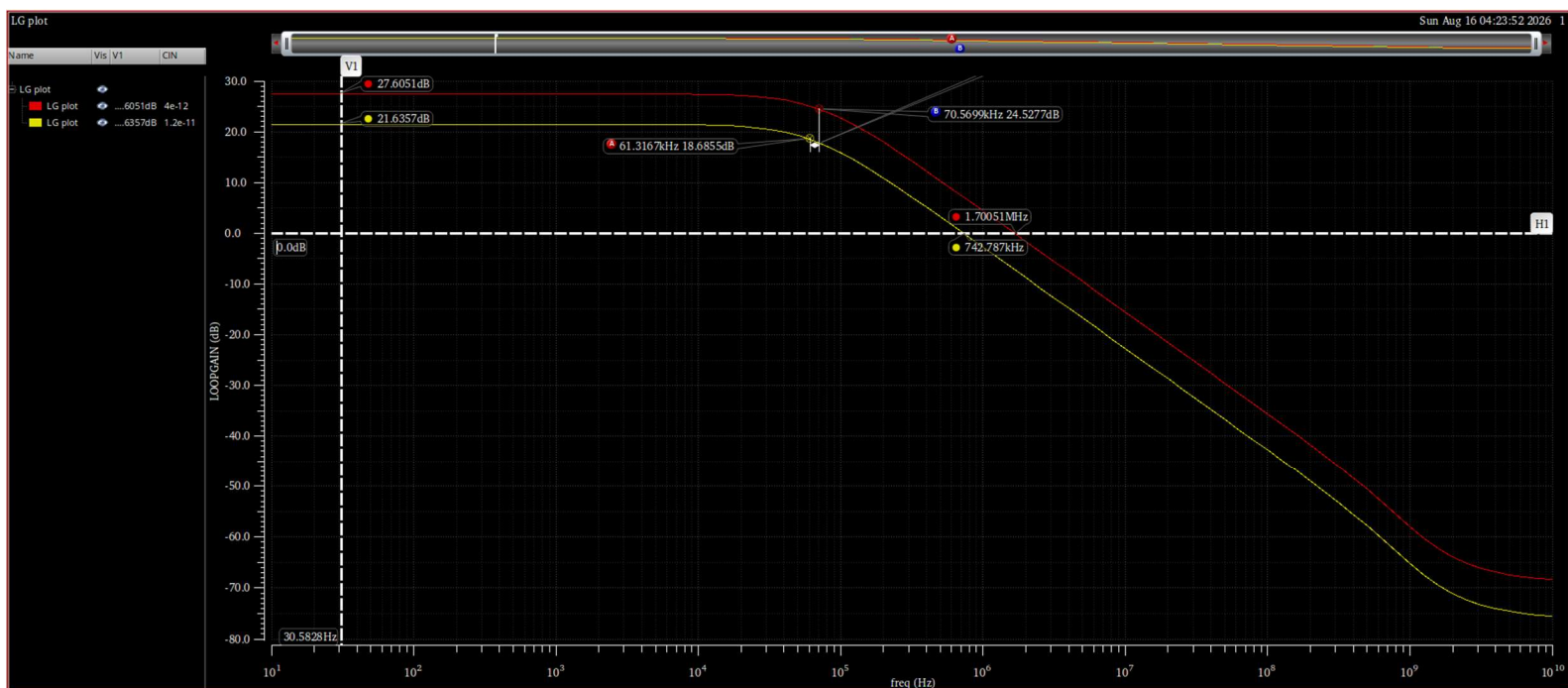
Point	Test	Output	Nominal
Parameters: CIN=4p			
1	lab8:TB:1	AVdb	5.65
1	lab8:TB:1	AV	1.917
1	lab8:TB:1	BW	1.767M
1	lab8:TB:1	GBW	3.395M
1	lab8:TB:1	VOUT	
Parameters: CIN=12p			
2	lab8:TB:1	AVdb	11.33
2	lab8:TB:1	AV	3.686
2	lab8:TB:1	BW	801.8k
2	lab8:TB:1	GBW	2.963M
2	lab8:TB:1	VOUT	



Comment

In the ideal behavioral circuit there was a buffer stage at the output so the C_{out} is not affected by the feedback and input caps as the increase the C_L which decreases the BW and GBW as $BW = 1 / (r_{out} * C_{out})$ but in the real ota circuit there weren't any buffer stage so the C_{out} was affected a lot and increased decreasing the BW and GBW.

Loop gain vs frequency



Point	Test	Output	Nominal
Parameters: CIN=4p			
1	lab8:TB:1	LG UGF	1.71M
1	lab8:TB:1	LG BW	70.59k
1	lab8:TB:1	LG	24
1	lab8:TB:1	LGdb	27.61
1	lab8:TB:1	LG plot	
Parameters: CIN=12p			
2	lab8:TB:1	LG UGF	746.8k
2	lab8:TB:1	LG BW	61.33k
2	lab8:TB:1	LG	12.07
2	lab8:TB:1	LGdb	21.64
2	lab8:TB:1	LG plot	



Comparison

CIN	Parameter	Part 1: Behavioral OTA	Part 2: Real 5T OTA
4 pF	Loop-gain UGF	2.502 MHz	1.710 MHz
4 pF	Loop-gain BW	97.47 kHz	70.59 kHz
4 pF	DC loop gain	25.62 V/V	24.00 V/V
4 pF	DC loop gain (dB)	28.17 dB	27.61 dB
12 pF	Loop-gain UGF	1.249 MHz	746.8 kHz
12 pF	Loop-gain BW	97.47 kHz	61.33 kHz
12 pF	DC loop gain	12.81 V/V	12.07 V/V
12 pF	DC loop gain (dB)	22.15 dB	21.64 dB

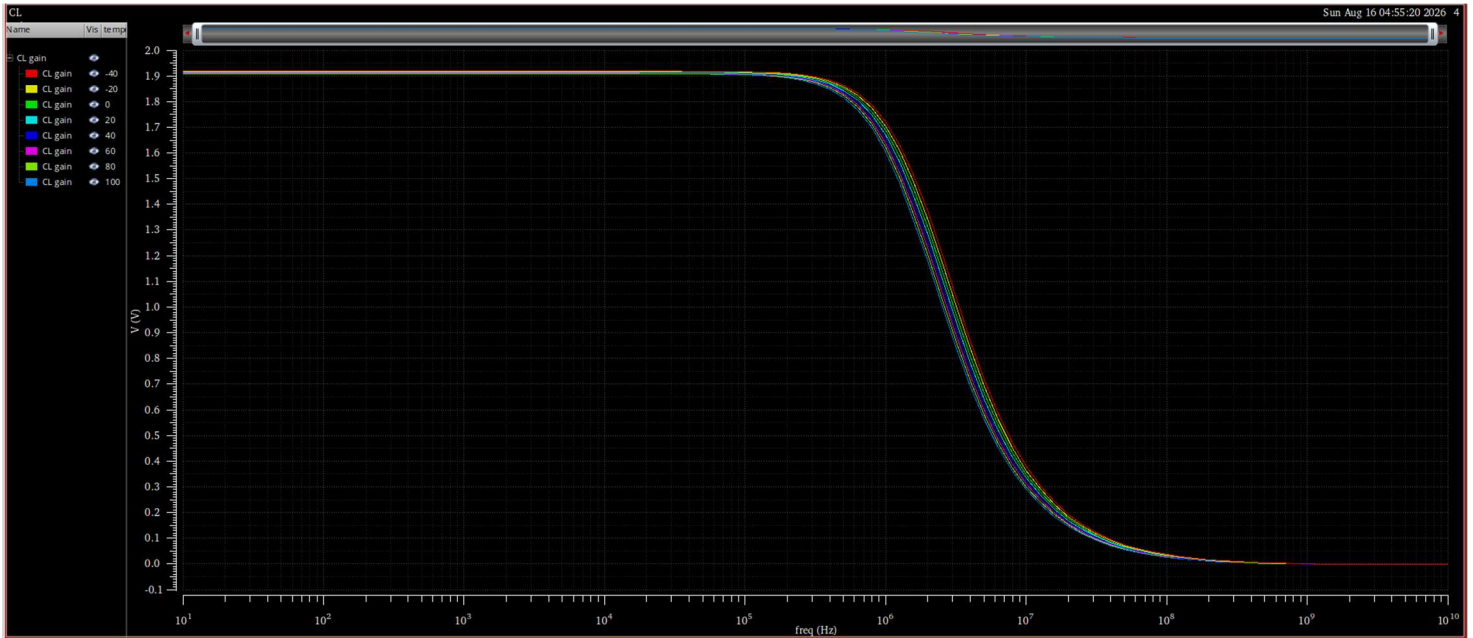
Comment

As $UGF = g_m / C_{out}$ and since C_{out} increased due to the load of the feedback and the C_{in} caps as they are parallel to the CL which increase the total C_{out} which decreases the UGF and the BW

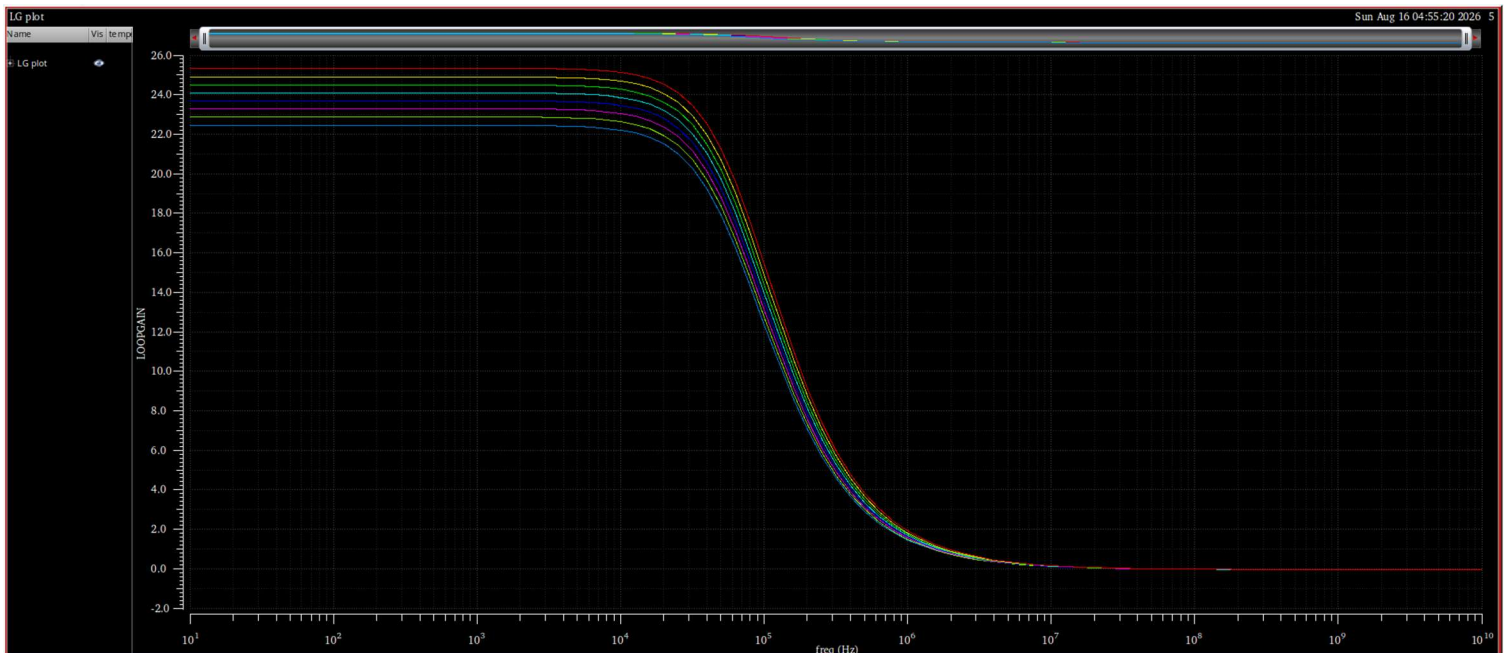


Gain Desensitization

Closed loop gain



Loop gain





Parameters: temperature=-40			
1	lab8:TB:1	CLAVdb	5.674
1	lab8:TB:1	CLAV	1.922
1	lab8:TB:1	CLBW	2.031M
1	lab8:TB:1	CLGBW	3.913M
1	lab8:TB:1	CLdbgain	
1	lab8:TB:1	LG UGF	1.96M
1	lab8:TB:1	LG BW	76.87k
1	lab8:TB:1	LG	25.39
1	lab8:TB:1	LGdb	28.09
1	lab8:TB:1	LG db plot	
1	lab8:TB:1	CL gain	
1	lab8:TB:1	LG plot	
Parameters: temperature=-20			
2	lab8:TB:1	CLAVdb	5.667
2	lab8:TB:1	CLAV	1.92
2	lab8:TB:1	CLBW	1.939M
2	lab8:TB:1	CLGBW	3.732M
2	lab8:TB:1	CLdbgain	
2	lab8:TB:1	LG UGF	1.882M
2	lab8:TB:1	LG BW	74.72k
2	lab8:TB:1	LG	24.95
2	lab8:TB:1	LGdb	27.94
2	lab8:TB:1	LG db plot	
2	lab8:TB:1	CL gain	
2	lab8:TB:1	LG plot	
Parameters: temperature=0			
3	lab8:TB:1	CLAVdb	5.66
3	lab8:TB:1	CLAV	1.919
3	lab8:TB:1	CLBW	1.862M
3	lab8:TB:1	CLGBW	3.58M
3	lab8:TB:1	CLdbgain	
3	lab8:TB:1	LG UGF	1.808M
3	lab8:TB:1	LG BW	72.82k
3	lab8:TB:1	LG	24.54
3	lab8:TB:1	LGdb	27.8
3	lab8:TB:1	LG db plot	
3	lab8:TB:1	CL gain	
3	lab8:TB:1	LG plot	

Parameters: temperature=20			
4	lab8:TB:1	CLAVdb	5.653
4	lab8:TB:1	CLAV	1.917
4	lab8:TB:1	CLBW	1.791M
4	lab8:TB:1	CLGBW	3.441M
4	lab8:TB:1	CLdbgain	
4	lab8:TB:1	LG UGF	1.735M
4	lab8:TB:1	LG BW	71.13k
4	lab8:TB:1	LG	24.14
4	lab8:TB:1	LGdb	27.66
4	lab8:TB:1	LG db plot	
4	lab8:TB:1	CL gain	
4	lab8:TB:1	LG plot	
Parameters: temperature=40			
5	lab8:TB:1	CLAVdb	5.646
5	lab8:TB:1	CLAV	1.916
5	lab8:TB:1	CLBW	1.725M
5	lab8:TB:1	CLGBW	3.313M
5	lab8:TB:1	CLdbgain	
5	lab8:TB:1	LG UGF	1.664M
5	lab8:TB:1	LG BW	69.63k
5	lab8:TB:1	LG	23.74
5	lab8:TB:1	LGdb	27.51
5	lab8:TB:1	LG db plot	
5	lab8:TB:1	CL gain	
5	lab8:TB:1	LG plot	
Parameters: temperature=60			
6	lab8:TB:1	CLAVdb	5.639
6	lab8:TB:1	CLAV	1.914
6	lab8:TB:1	CLBW	1.663M
6	lab8:TB:1	CLGBW	3.192M
6	lab8:TB:1	CLdbgain	
6	lab8:TB:1	LG UGF	1.593M
6	lab8:TB:1	LG BW	68.29k
6	lab8:TB:1	LG	23.34
6	lab8:TB:1	LGdb	27.36
6	lab8:TB:1	LG db plot	
6	lab8:TB:1	CL gain	
6	lab8:TB:1	LG plot	



Compare the percent change

loop gain

at -40=25.39

at 100=22.5

change is

$$\frac{22.5-25.39}{25.39} \times 100 = -11.38\%$$

Closed loop gain

at -40=1.922

at 100=1.911

change is

$$\frac{1.911-1.922}{1.922} \times 100 = -0.5723\%$$

Comment

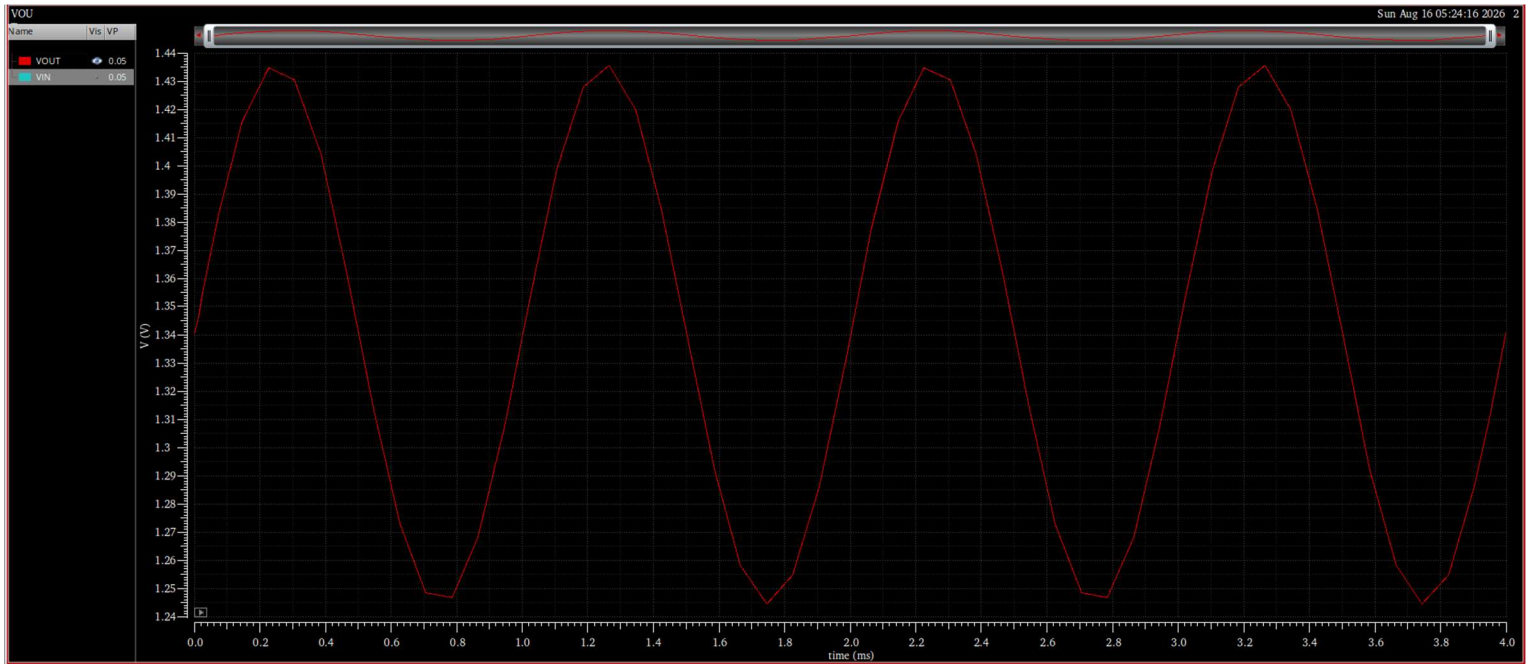
The closed loop gain change is a lot less due to the Desensitization of the gain as in the Acl once the LG >>1 then we can approximate that Acl=1/beta so it is less sensitive to variations also it decrease as the mobility decreases with rising temps reducing the GM.

Parameters: temperature=80			
7	lab8:TB:1	CLAVdb	5.632
7	lab8:TB:1	CLAV	1.912
7	lab8:TB:1	CLBW	1.605M
7	lab8:TB:1	CLGBW	3.076M
7	lab8:TB:1	CLdbgain	
7	lab8:TB:1	LG UGF	1.545M
7	lab8:TB:1	LG BW	67.11k
7	lab8:TB:1	LG	22.92
7	lab8:TB:1	LGdb	27.21
7	lab8:TB:1	LG db plot	
7	lab8:TB:1	CL gain	
7	lab8:TB:1	LG plot	
Parameters: temperature=100			
8	lab8:TB:1	CLAVdb	5.624
8	lab8:TB:1	CLAV	1.911
8	lab8:TB:1	CLBW	1.552M
8	lab8:TB:1	CLGBW	2.973M
8	lab8:TB:1	CLdbgain	
8	lab8:TB:1	LG UGF	1.5M
8	lab8:TB:1	LG BW	66.07k
8	lab8:TB:1	LG	22.5
8	lab8:TB:1	LGdb	27.04
8	lab8:TB:1	LG db plot	
8	lab8:TB:1	CL gain	
8	lab8:TB:1	LG plot	

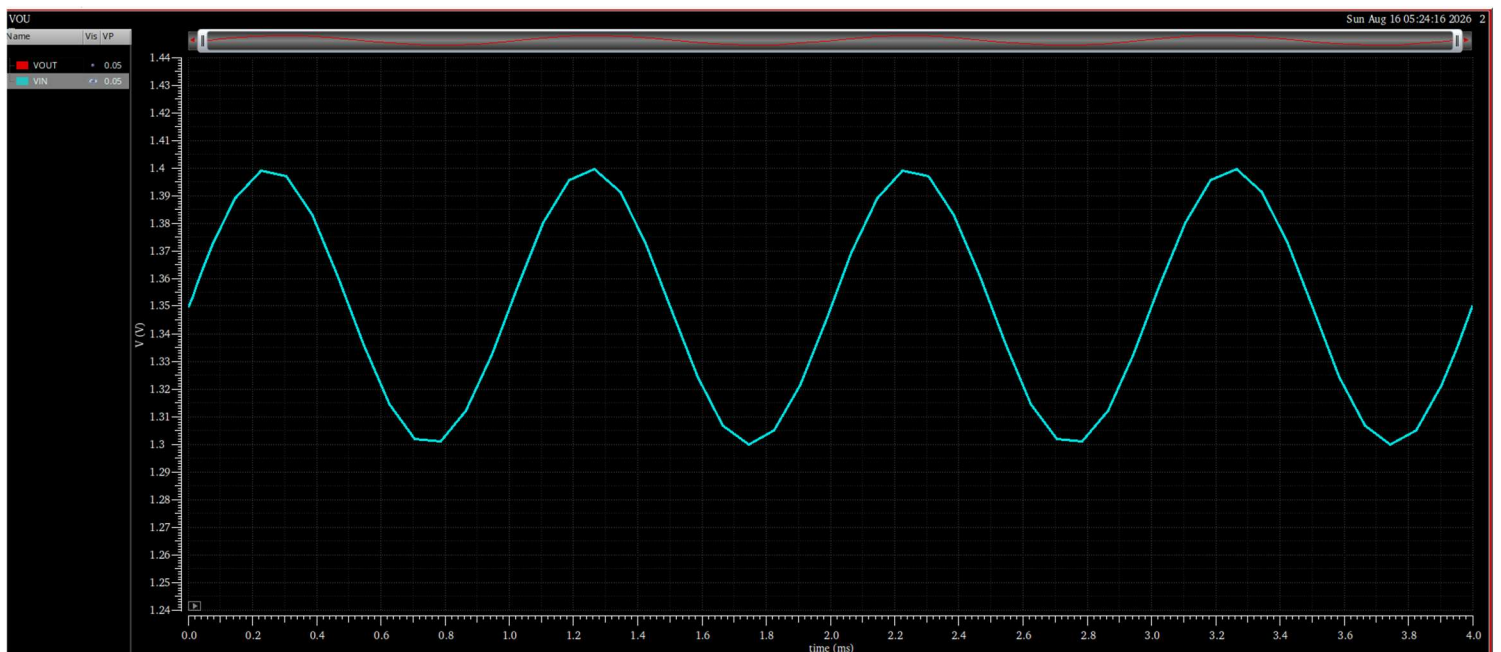


Transient analysis

VOUT at 1kHz

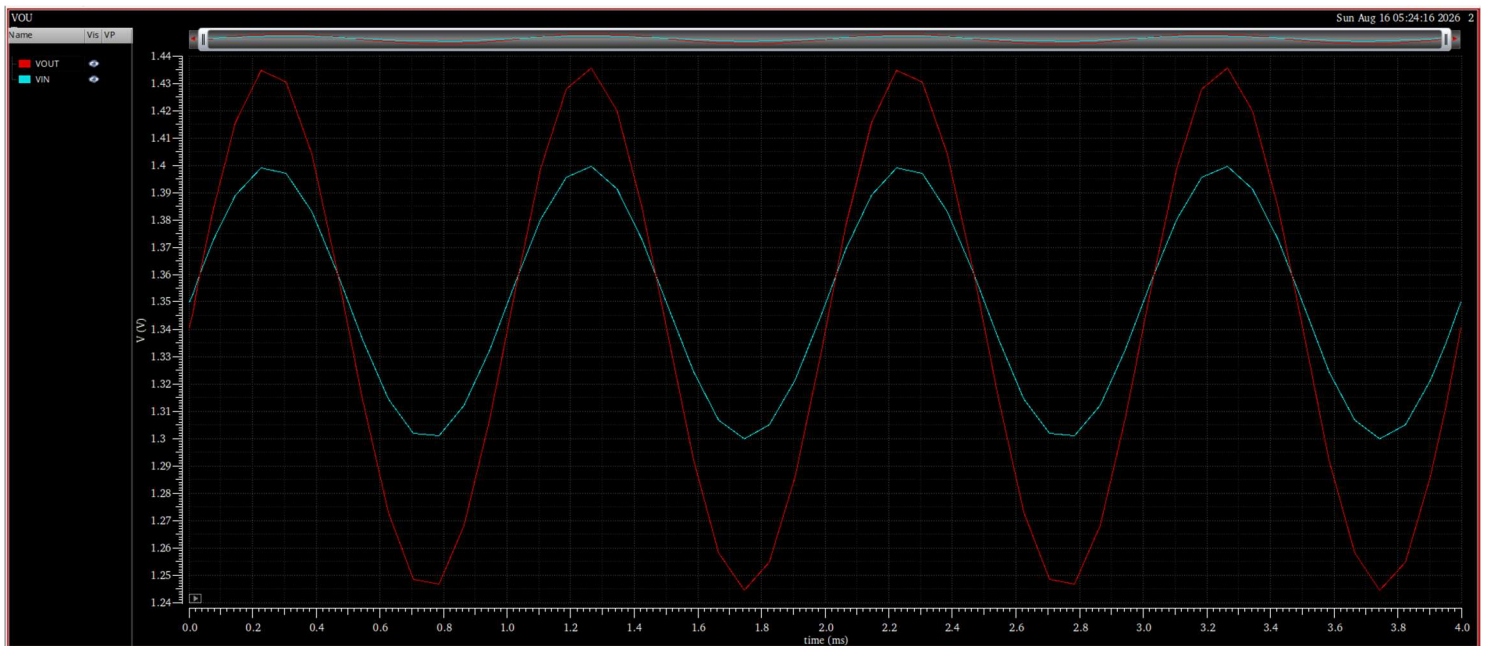


Vin at 1kHz

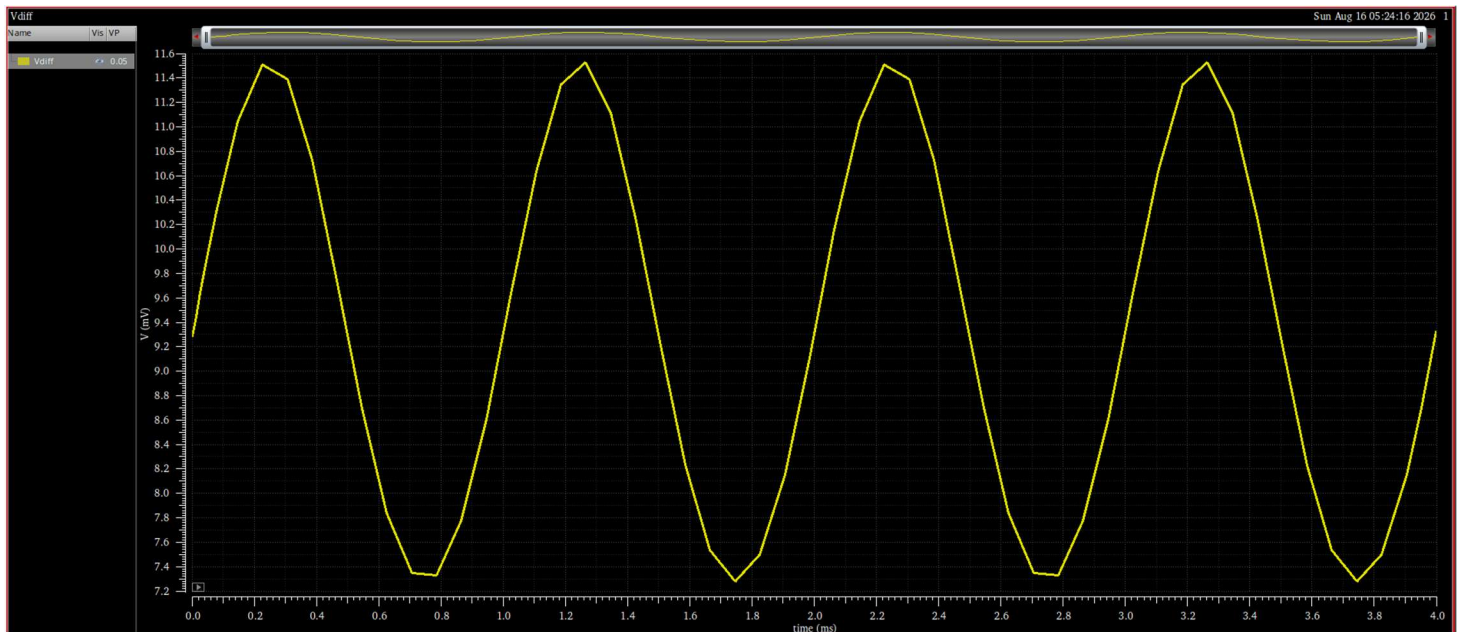




Vin and Vout overlaid at 1kHz



Vdiff at 1kHz





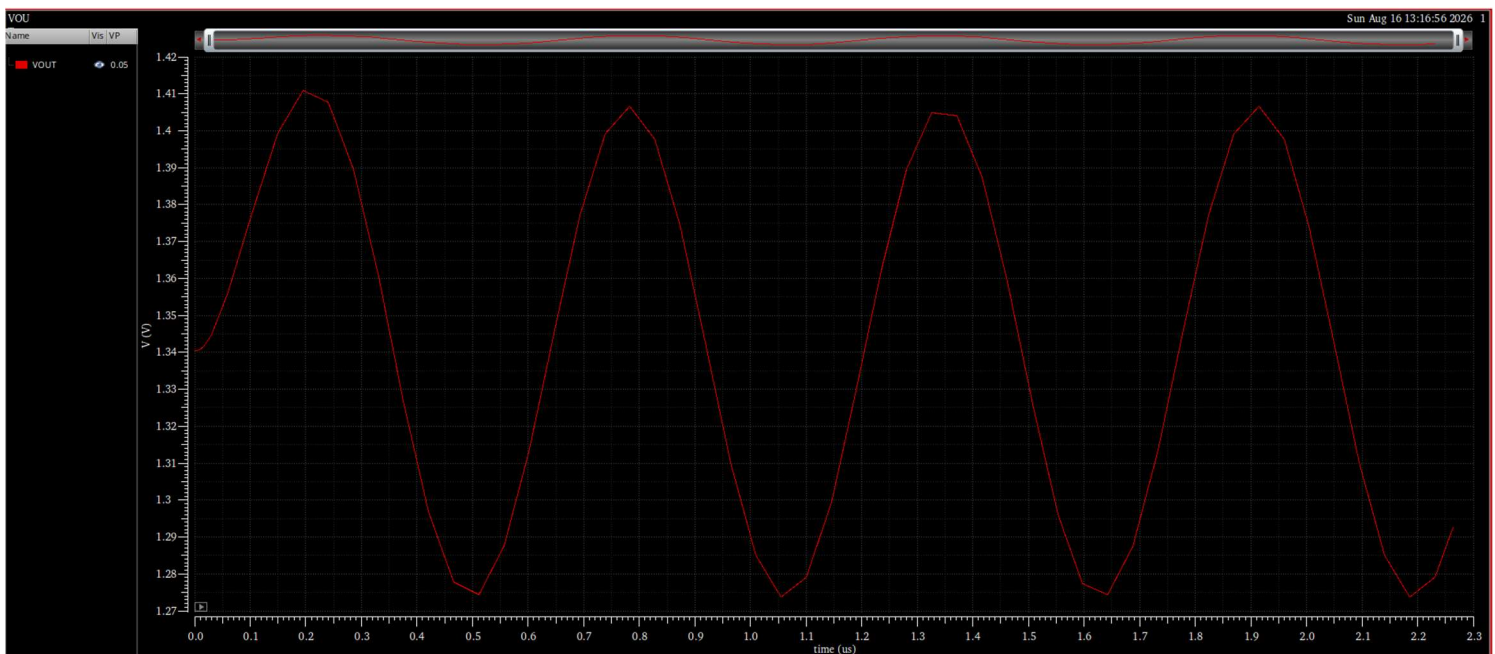
Peak to Peak voltages at 1kHz

lab8:TB:1	VINpp	99.79m
lab8:TB:1	VOUTpp	191.2m
lab8:TB:1	Vdiffpp	4.254m

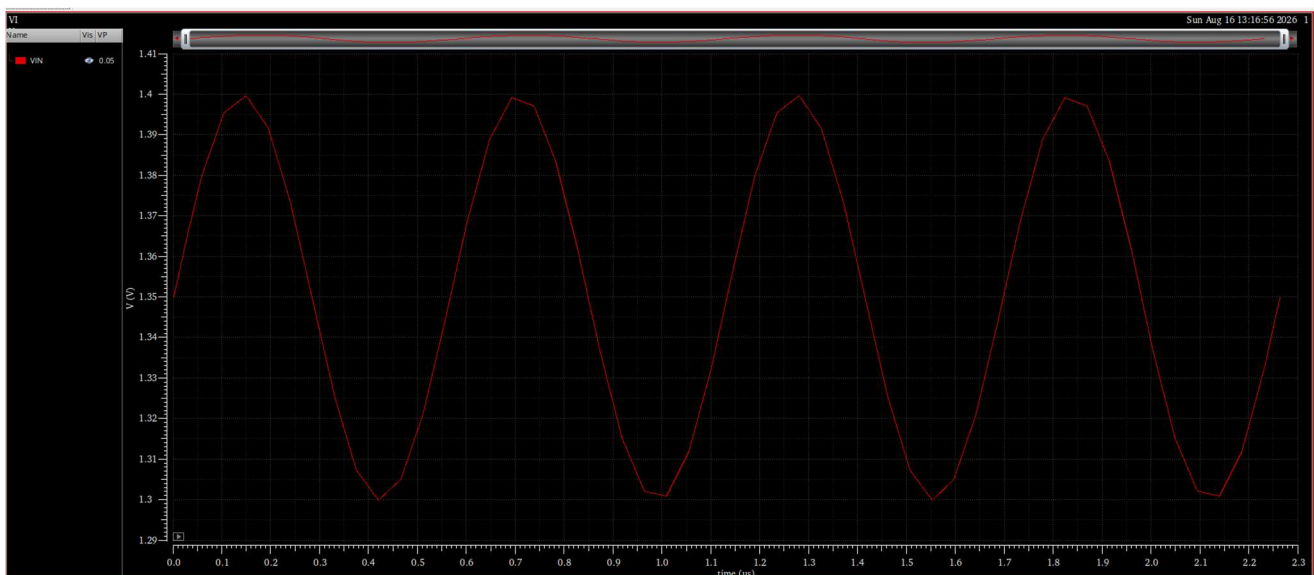
relation between the output and (VP – VN)

The open loop gain is the relation between the the (Vp-Vn) and the Vout as $V_{out} = A_{ol} * (V_p - V_n)$ so that's why Vdiff is small

Vout at 1.767MHz

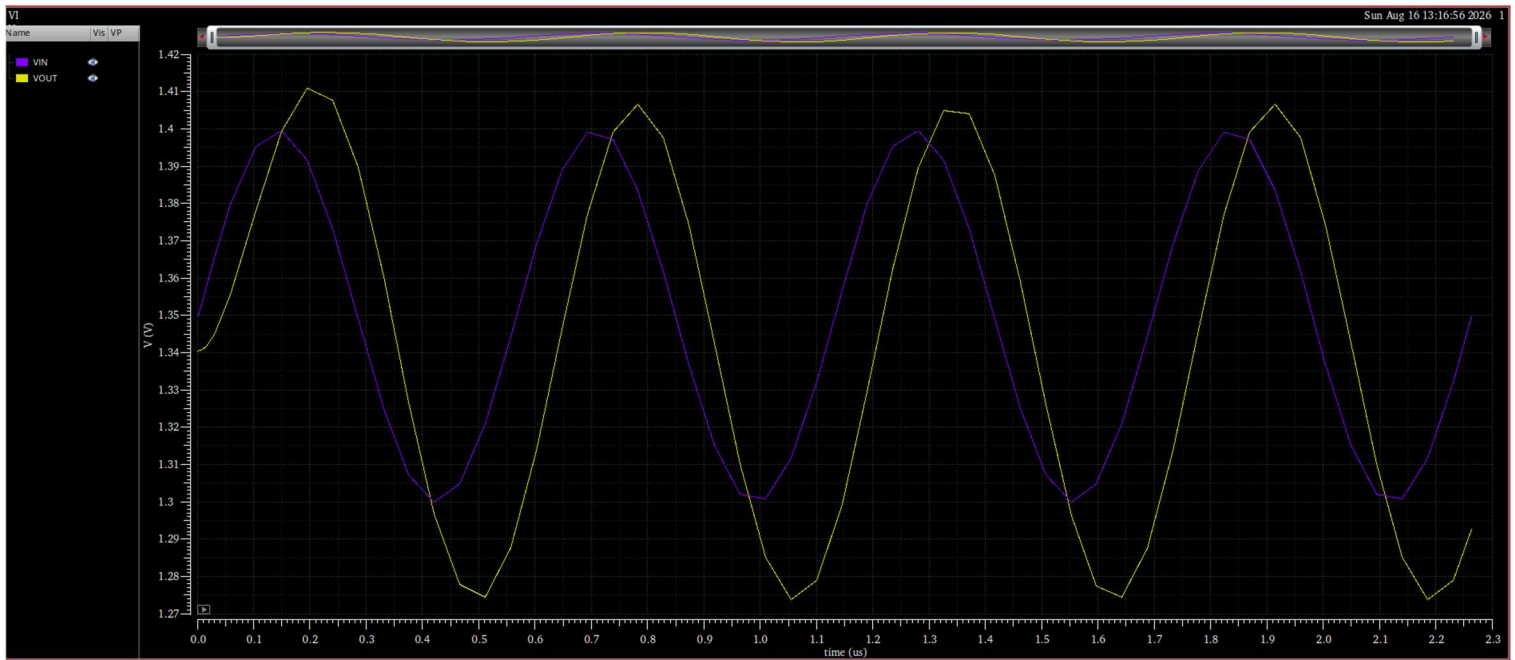


VIN at 1.767MHz

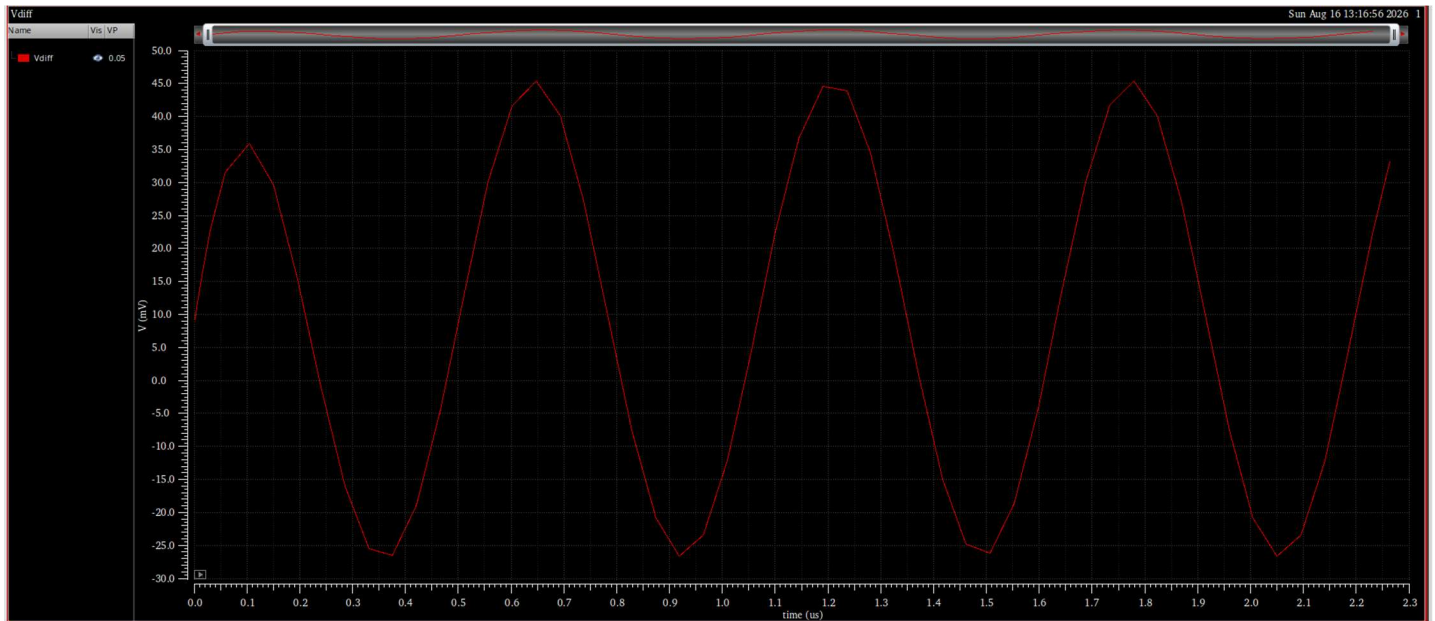




Vin and Vout overlaid at 1.767MHz



Vdiff at 1.767MHz





Peak to Peak voltages at 1.767MHz

lab8:TB:1	VINpp	99.79m
lab8:TB:1	VOUThp	137.1m
lab8:TB:1	Vdiffhp	71.97m

The relation between the output and the input signal

It is the closed loop gain which is equal $A_{ol}/1+LG$ which approximates to $1/\beta$ if $LG \gg 1$,

The relation between the output and (VP – VN)

It is the open loop gain

Compare

At low frequency in the first case the A_{ol} did not start to decrease at the V_{diff} was relatively small as there is a large A_{ol} and the static error was also small due to large LG while in the second case at a high frequency = to the BW closed loop the A_{ol} start to decrease sharply due to the frequency and apparently it became similar to the A_{cl}

$$A_{OL}(s) = \frac{A_0}{1 + \frac{s}{\omega_p}}$$

$$|A_{OL}(j\omega)| = \frac{A_0}{\sqrt{1 + \left(\frac{\omega}{\omega_p}\right)^2}} \approx \frac{A_0 \omega_p}{\omega}$$

$$\omega_u \approx A_0 \omega_p$$

$$|A_{OL}(j\omega)| \approx \frac{\omega_u}{\omega}$$

So from this equation we can state that BWcl the A_{ol} is equal to the $A_{cl} = 1/\beta$

Due to this decrease the static error increases and the V_{diff} increase .

At 1.767MHz

lab8:TB:1	VINpp	99.79m
lab8:TB:1	VOUThp	137.1m
lab8:TB:1	Vdiffhp	71.97m

At 1KHz

lab8:TB:1	VINpp	99.79m
lab8:TB:1	VOUThp	191.2m
lab8:TB:1	Vdiffhp	4.254m